NeRI: Implicit Neural Representation of LiDAR Point Cloud using Range Image Sequence

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The code is accessible in https://github.com/RuixiangXue/NeRI

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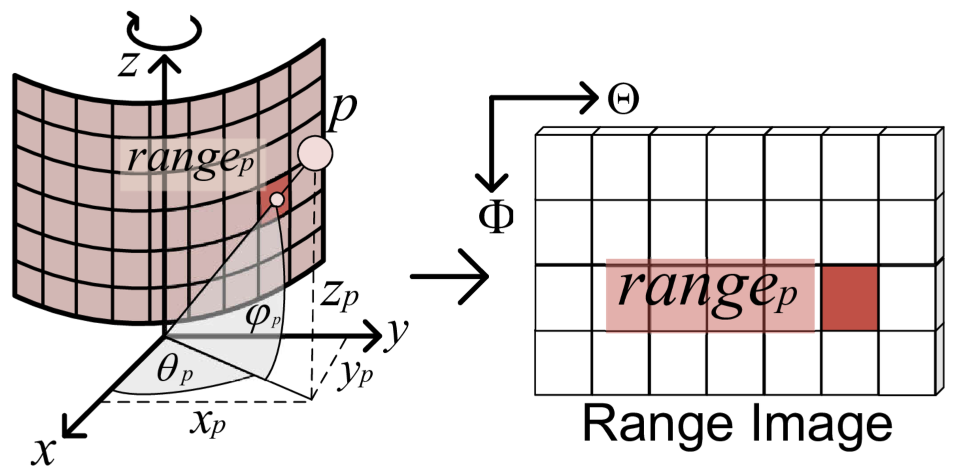
Introduction

- This paper proposes the *NeRI*, an implicit neural representation (INR) based LiDAR point cloud compressor. In *NeRI*, we first transform a sequence of 3D LiDAR frames into a 2D range image sequence through range image projection over time. Then, we employ a neural network conditioned on the temporal frame index and associated LiDAR sensor pose to fit input range images as closely as possible. The optimized network parameters, which implicitly represent the input LiDAR data, are later lossily compressed. *NeRI* decoder is then initialized using decoded parameters to generate range images for reconstructing the 3D LiDAR sequence accordingly. Extensive experimental results demonstrate the significant superiority of *NeRI* regarding the compression efficiency and decoding speed compared to state-of-the-art 2D and 3D compressors for LiDAR point cloud.

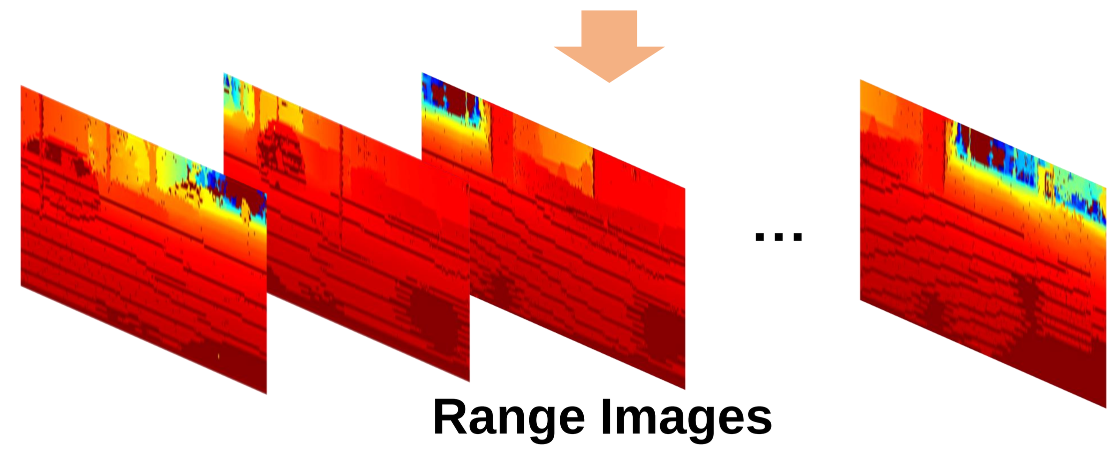
Range image Representation



$$\begin{cases} range_p = \sqrt{x_p^2 + y_p^2 + z_p^2} \\ \theta_p = arctan(x_p/y_p) \\ \phi_p = arcsin(z_p/range_p) \end{cases}$$

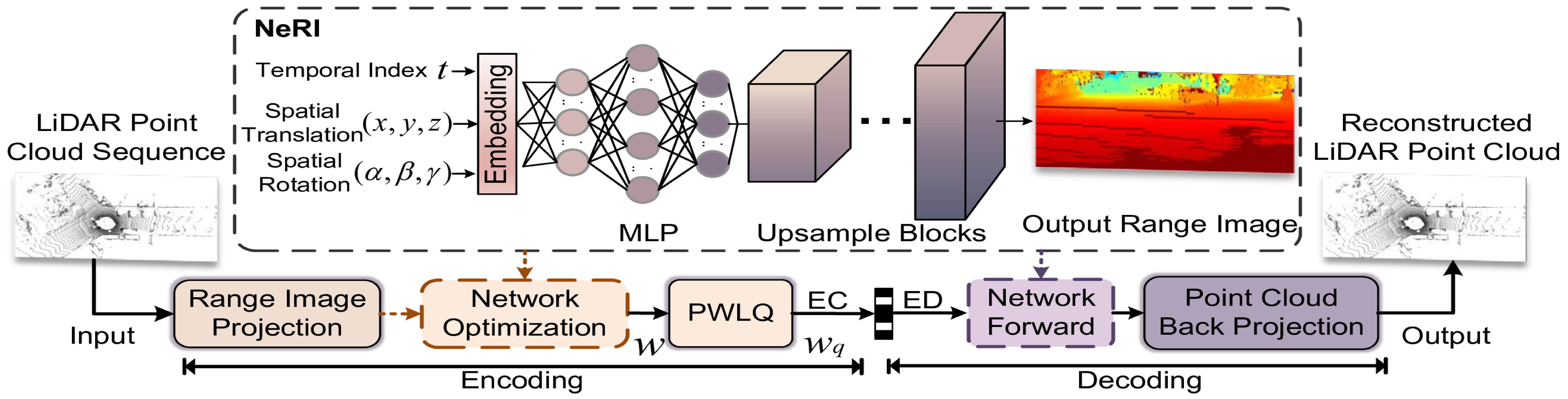
$$\begin{cases} u_p = [\frac{\theta_p \cdot w}{FOV_h}] \\ v_p = [(\phi_p - \phi_{min})/(\frac{\phi_{max} - \phi_{min}}{h-1})] \end{cases}$$

Range Image Projection



Range Images

Implicit Neural Representation of LiDAR Point Cloud



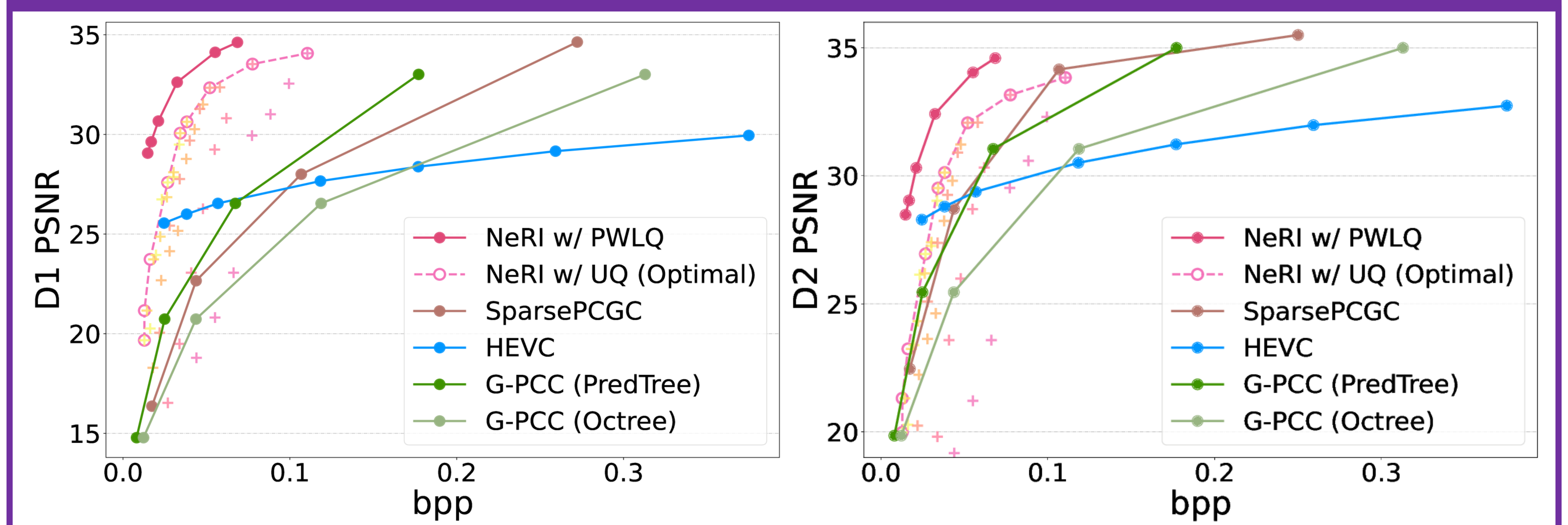
- The overall quantization range of model parameters $[-m, m]$ is separated into two symmetric regions.

$$\mathrm{PWLQ}(w; b, m, p) = \begin{cases} sign(w) \cdot UQ(|w|; b-1, 0, p, 0), w \in R_1 \\ sign(w) \cdot UQ(|w|; b-1, p, m, p), w \in R_2 \end{cases}$$

- We formulate the input of *NeRI* as three parts **Timestamp** t, **Spatial translation** T_t, and **Spatial Rotation** R_t.
- We iteratively train and optimize the network in a self-supervised manner.

$$\mathcal{L}_1 = \frac{1}{T} \sum_i |y_t - f_\theta(\mathbf{x_t})|$$

Compression Performance



Rate-Distortion Curves (D1 and D2 PSNR)

Seq.	Number of Frames	BD-Rate (D1)			BD-Rate (D2)		
		G-PCC (PredTree)	HEVC	SparsePCGC	G-PCC (PredTree)	HEVC	SparsePCGC
#00	4541	-82.90%	-94.59%	-84.96%	-62.88%	-80.84%	-49.80%
#02	4660	-94.20%	-93.18%	-83.28%	-58.45%	-81.45%	-50.16%
#08	4071	-78.31%	-86.16%	-68.55%	-42.93%	-85.17%	-52.33%
Avg.	**4424**	**-85.14%**	**-91.31%**	**-78.93%**	**-54.75%**	**-82.48%**	**-50.76%**

BD-BR Gain over G-PCC

- We conduct LiDAR Point cloud compression using *NeRI* on **KITTI**.
- We compare the compression performance with advanced compressor, e.g., **G-PCC**, **HEVC**, and **SparsePCGC**.

Decoding Speed

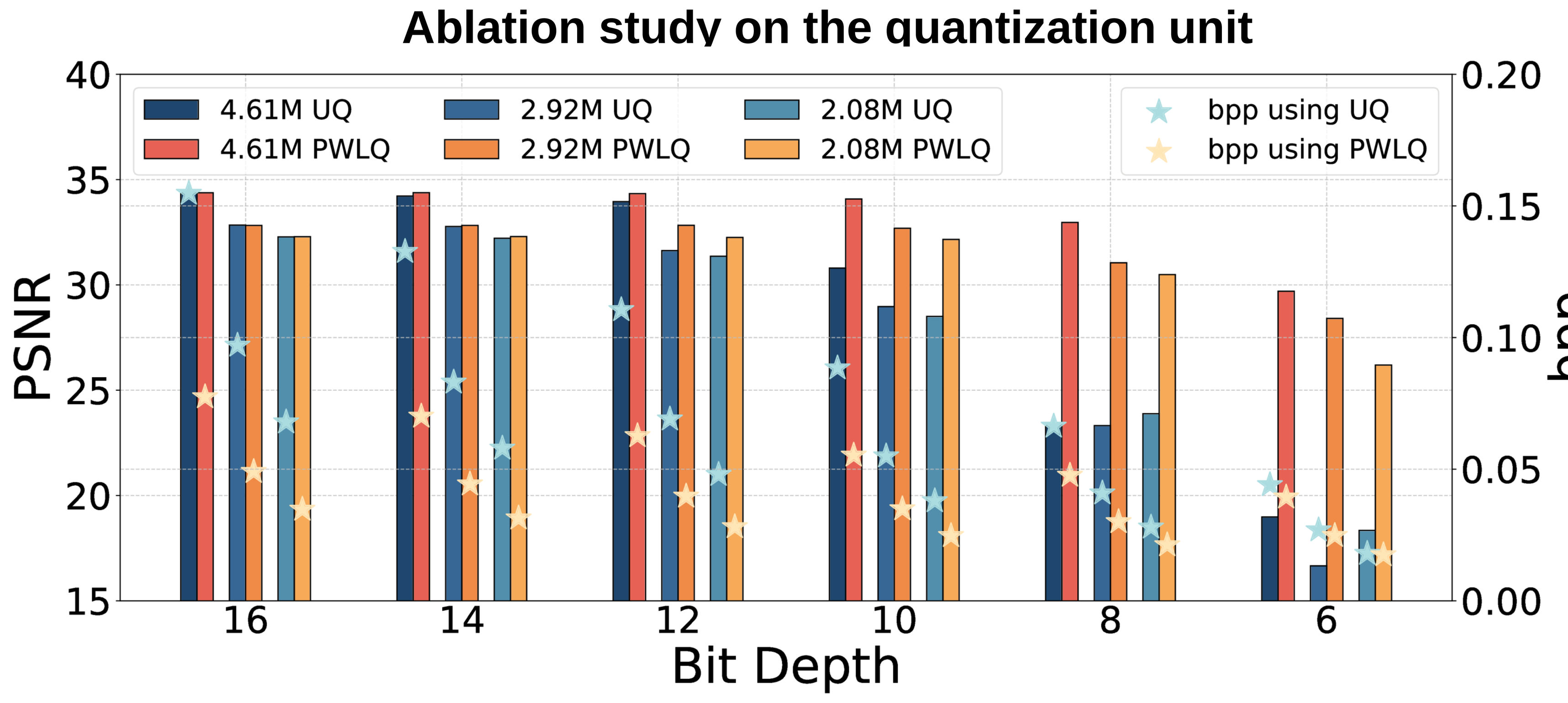
- We test the decoding speed of different methods on sequences #00, #02, and #08 of KITTI and report their average **FPS (frames per second).**

Method	FPS ↑
G-PCC (Octree)	9.68
G-PCC (PredTree)	11.85
HEVC	107.86
SparsePCGC	1.65
NeRI	**376.39**

Quantization

- We compare the compression performance of **Uniform Quantization (UQ)** and **Piece-wise Linear Quantization (PWLQ)** under different bit depths and parameter sizes.

Ablation study on the quantization unit



Positional Embedding

- We then examine the impact of allocating varying lengths to individual items.
- The assignment of **40/60/60** to $\{t, T_t, R_t\}$ attains the highest PSNR performance.

Ablation study on the positional embedding

	Embedding Length of $\{t, T_t, R_t\}$			
	160/0/0	80/40/40	40/60/60	0/80/80
D1 PSNR	42.24	42.54	**42.73**	42.46
D2 PSNR	43.10	43.20	**43.24**	43.14